

Automated segmentation of the left ventricle in cardiac MRI

Michael R. Kaus^{a,*}, Jens von Berg^a, Jürgen Weese^a, Wiro Niessen^b, Vladimir Pekar^a

^a Philips Research Laboratories, Sector Technical Systems, Röntgenstr. 24-26, D-22335 Hamburg, Germany

^b Image Sciences Institute, University Medical Center, Heidelberglaan 100, 3584 CX Utrecht, The Netherlands

Available online 31 July 2004

Abstract

We present a fully automated deformable model technique for myocardium segmentation in 3D MRI. Loss of signal due to blood flow, partial volume effects and significant variation of surface grey value appearance make this a difficult problem. We integrate various sources of prior knowledge learned from annotated image data into a deformable model. Inter-individual shape variation is represented by a statistical point distribution model, and the spatial relationship of the epi- and endocardium is modeled by adapting two coupled triangular surface meshes. To robustly accommodate variation of grey value appearance around the myocardial surface, a prior parametric spatially varying feature model is established by classification of grey value surface profiles. Quantitative validation of 121 3D MRI datasets in end-diastolic (end-systolic) phase demonstrates accuracy and robustness, with 2.45 mm (2.84 mm) mean deviation from manual segmentation.

© 2004 Elsevier B.V. All rights reserved.

Keywords: Image post-processing; Segmentation; Deformable models; Shape models; Cardiac MRI

1. Introduction

Cardiac MRI provides important information for diagnosis and treatment of cardiovascular diseases by enabling quantitative assessment of functional parameters such as wall motion, wall thickness and ejection fraction (Frangi et al., 2001). However, the lack of computational tools still implies that not all the information contained in the data is currently used. In order to fully exploit the information in the MRI data in clinical practice, and considering the amount of data involved (10–20 frames by 8–12 image slices), automated extraction of the myocardium is essential. However, poor object features resulting from signal loss due to blood flow and partial volume effects, and significant variation of grey value appearance make finding a robust and accurate solution a difficult problem.

Recent approaches to automated segmentation incorporate prior knowledge to improve robustness (Jolly et al., 2001; Lorenzo-Valdes et al., 2002; Mitchell et al., 2001; Montagnat and Delingette, 2000). Lorenzo-Valdes et al. (2002) achieved convincing results by volumetric atlas matching using B-spline registration. However, this approach requires several minutes for a single time frame. The fast and robust framework of active shape and appearance models by Cootes et al. (1998) was extended and applied to cardiac 3D and 2D time series (Mitchell et al., 2001), achieving promising results. However, statistical shape models cannot capture variability outside the learning set, which is likely to occur in the case of pathology, and appearance modeling may fail in the presence of large grey value variability across subjects and time. Learning sets could be used for specific targeted pathologies, but this is time consuming and requires a large database of pathologic cases.

Earlier work by this group proposed shape constrained deformable models, enabling local deviation

* Corresponding author.

E-mail address: michael.kaus@philips.com (M.R. Kaus).

from a statistical shape model by embedding it into a 3D triangular deformable mesh (Kaus et al., 2003; Weese et al., 2001). This approach used deterministic feature search and was successfully applied to segment bones in CT. However, like many other deformable model approaches (McInerney and Terzopoulos, 1996), multiple objects like, e.g., the endo- and epicardium are not considered, or objects of significant shape variation over time and across subjects. In addition, methods with deterministic feature functions assume constant grey value appearance of the object's surface, which is not true for the myocardial surface due to different surrounding tissue as, e.g., lung parenchyma, fat, and the blood pool in the right ventricle.

We propose to integrate several prior information sources into deformable models including a deterministic, parametric model of the variation of surface features, inter- and intra-subject shape variation, and spatial relationships to handle multiple objects. In the remainder of this article we present the modeling and segmentation methods with application to cardiac 3D MRI time series.

2. Methods

In this section we outline the deformable model framework (Weese et al., 2001) and then describe how to represent, learn and integrate prior knowledge into the method.

2.1. Deformable surface models

Our approach is closely related to the deformable models pioneered by Kass et al. (1988) and related to the work of Montagnat and Delingette (1998). The deformable model is represented by a triangular mesh. After initial positioning, the mesh is adapted to the target surface in a 3D image by iteratively carrying out surface detection in the image for each triangle, and reconfiguration of the mesh vertices. Mesh reconfiguration is done by minimizing

$$E = E_{\text{ext}} + \alpha E_{\text{int}}. \quad (1)$$

The external energy E_{ext} drives the mesh towards the surface points obtained in the surface detection step. The internal energy E_{int} restricts the flexibility by maintaining the vertex configuration of an initial mesh. The parameter α weighs the relative influence of each term. These steps are repeated a fixed number of iterations i . The different components of the deformation algorithm are described in the following.

2.1.1. Surface detection

Surface detection is carried out for each triangle barycenter $\mathbf{x}_i$. We seek the point $\tilde{\mathbf{x}}_i$ along the triangle normal

$\mathbf{n}_i$ which maximizes the objective function consisting of a feature function F and distance to the triangle barycenter according to

$$\tilde{\mathbf{x}}_i = \mathbf{x}_i + kh\mathbf{n}_i, k = \arg \max_{j=-l, \dots, l} \{F(\mathbf{x}_i + jh\mathbf{n}_i) - dh^2j^2\}, \quad (2)$$

where $2l+1$ is the number of points investigated along the search profile, h specifies the distance between two points on the profile, and d controls the tradeoff between feature strength and distance.

The feature function F is designed to detect image acquisition characteristics specific to the object's surface (Section 2.2.3). A simple example would be the image gradient.

By combining feature value and distance information, surface points next to the mesh are preferred and mesh adaptation becomes more stable. If we would search for the point with the maximum feature value ($d=0$), the parameter l became a crucial parameter. For small l , proper surface points could only be detected for a rather accurate initialization. If l were large, false object boundaries are detected rather frequently. Thus the search range is set large enough to capture points in the image, but as small as possible in order to limit the computation time. The distance between two profile points is set in accordance with the image resolution.

2.1.2. External energy

The external energy term drives the triangle barycenters $\mathbf{x}_i$ of the mesh towards the detected surface points

$$E_{\text{ext}} = \sum_{i=1}^T w_i (\mathbf{e}_{\nabla I}(\tilde{\mathbf{x}}_i - \mathbf{x}_i))^2, \quad w_i = \max\{0, F(\mathbf{x}_i) - h^2j^2d\}, \quad (3)$$

where T is the number of triangles, $\mathbf{x}_i$ is the coordinate of the triangle barycenter to be computed, and $\mathbf{e}_{\nabla I}$ is the unit vector in the direction of the image gradient at the feature point $\tilde{\mathbf{x}}_i$.

Projection onto $\mathbf{e}_{\nabla I}$ allows the triangle center to slide along a surface patch associated with the feature point (Fig. 1). It has been demonstrated that this strategy reduces the risk of trapping by false structures compared

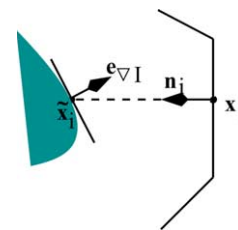


Fig. 1. Feature search through a triangle's barycenter. The triangle center is attracted by the surface patch associated with the detected feature point $\tilde{\mathbf{x}}_i$. This reduces the risk of trapping by false features.

to directly attracting the triangle centers by the feature point (Weese et al., 2001). In the latter case, triangle centers that have been attracted by false feature points, which are detected frequently at the beginning of the adaptation process, could almost not move anymore. To enable a proper distribution of the mesh vertices, the internal force would therefore have to be very large. A large internal force restricts, however, the flexibility of the mesh and degrades the adaptation accuracy.

The weights w_i are defined in accordance with the feature detection objective function (2) and give the most promising surface points $\tilde{x}_i$ the largest influence during mesh reconfiguration. This also slightly reduces the necessary number of iterations.

2.1.3. Internal energy

The internal energy maintains the distribution of the vertex coordinates v_j and is defined as

$$E_{\text{int}} = \sum_{j=1}^V \sum_{k \in N(j)} (v_j - v_k - s\mathbf{R}(v_j^1 - v_k^1))^2, \quad (4)$$

where $N(j)$ is the set of neighbours of vertex j , v_j^1 are the vertices of the initial mesh, and V is the number of vertex coordinates. The neighbouring vertices are those connected by a single triangle edge.

In the case of clear cut outliers in the segmentation, the internal energy and the given fixed template mesh topology, which is a part of our correspondence model, will prevent the model to deform towards the outliers. Explicit estimation of the scaling s and the rotation $\mathbf{R}$ that align the deforming with the initial mesh makes the internal energy invariant to pose and scaling of the target structure in the image.

2.1.4. Optimization

The energy (1) is minimized in a two-step optimization procedure. First, the orientation $\mathbf{R}$ and the scaling s are estimated with a fast closed-form point-based registration method based on SVD (Golub and van Loan, 1996). Given $\mathbf{R}$ and s , we estimate the vertex coordinates. The quadratic form of the energies (3), (4) is important since it ensures fast convergence. Due to the locality of the connections between mesh vertices, energy minimization results in solving a sparse linear equation system, which can be done efficiently using the conjugate gradient method (Gill et al., 1981).

2.2. Integration of multi-model prior knowledge

In this section we explain how multi-model information can be efficiently integrated into a deformable shape model, i.e. a model of inter- and intra-subject shape variability, a deterministic, parametric model of the variation of surface features, and spatial relationships to handle multiple objects.

2.2.1. Prior model of shape variability

2.2.1.1. Point distribution models. A point distribution model (PDM) is a parametric representation of a set of learning shapes, where each shape is represented by a set of labeled landmark points (Cootes et al., 1998). An arbitrary shape $\mathbf{s} = [\mathbf{v}_1^T, \dots, \mathbf{v}_V^T]^T$, where $\mathbf{v}_i = [v_{ix} \ v_{iy} \ v_{iz}]^T$ is a landmark coordinate, is approximated by the sum of the mean shape $\mathbf{m}$ and $L-1$ weighted eigenmodes $\mathbf{e}_l$,

$$\mathbf{s} \approx \mathbf{m} + \sum_{l=1}^{L-1} p_l \mathbf{e}_l, \quad p_l = (\mathbf{s} - \mathbf{m})^T \mathbf{e}_l. \quad (5)$$

The PDM is generated from the set of shape vectors by principal component analysis (PCA), where the eigenmodes are the eigenvectors of the cross-correlation matrix $\mathbf{C}$ capturing deviations from the mean shape

$$\mathbf{m} = \frac{1}{L} \sum_{i=1}^L \mathbf{s}_i, \quad \mathbf{C} = \frac{1}{L-1} \sum_{i=1}^L (\mathbf{s}_i - \mathbf{m})(\mathbf{s}_i - \mathbf{m})^T, \quad (6)$$

where L is the number of learning shapes $\mathbf{s}_i$.

2.2.1.2. PDM construction. One of the most crucial aspects of PDM construction is the generation of anatomical landmarks. Due to the large number of corresponding landmarks required, manual indication is not feasible. Our group has previously described an automated method to generate 3D statistical shape models from a set of L segmented images (Kaus et al., 2003) which is outlined below.

For each segmented learning image I_i , we seek a triangular mesh with vertices $\{\mathbf{v}_j^i \mid j = 1, \dots, V\}$, where the j th vertex corresponds to approximately the same surface location from one shape to another. A consistent triangulation of all objects in the learning set is constructed by first selecting an arbitrary template from the set of learning images, and triangulating the object surface such that the vertex density is proportional to the surface curvature (Krahnstöver and Lorenz, 2004). In this way we derive a mesh with as few vertices and triangles as possible to minimize the computation time during mesh adaptation.

Subsequently, the template mesh is adapted to the remaining shapes using the deformable model approach described above, where the feature function is the projected image gradient $F = \mathbf{n}^T \mathbf{g}$. This procedure yields a set of triangulated surfaces with pointwise, approximately anatomically corresponding vertices, which are aligned by Procrustes analysis before estimation of the PDM. The choice of the template may affect the number of vertices and triangles, but has no effect on the accuracy of the model.

2.2.1.3. Integration into deformable surface adaptation. A PDM can be adapted to unseen image data by finding the optimal weights p_i (Cootes et al., 1998), but shapes

outside the variation of the training set cannot be described. Cootes and Taylor (1995) provide additional variability by combining the eigenmodes learned from shape examples and artificial vibration modes from a linear physically based finite element model (Pentland and Sclaroff, 1991). This combined formulation was shown to perform better than either of the mentioned methods alone, but at the cost of computationally intensive finite element analysis to solve for the modes of vibration.

We propose a simpler approach which allows local deviations from the trained shape space (Weese et al., 2001). The PDM is embedded into the deformable model by encouraging the distribution of the mesh vertices to stay close to the shape space spanned by the PDM by extending the internal energy (4) to

$$E_{\text{int}} = \sum_{j=1}^V \sum_{k \in N(j)} \left(\mathbf{v}_j - \mathbf{v}_k - s\mathbf{R} \left(\mathbf{m}_j - \mathbf{m}_k + \sum_{l=1}^{\tilde{L}} p_l (\mathbf{e}_{lj} - \mathbf{e}_{lk}) \right) \right)^2, \quad (7)$$

where $\tilde{L}$ is the number of eigenmodes used, $\mathbf{m}_j$, $\mathbf{e}_{lj}$ are the vertices of the PDM, and the p_l are calculated each iteration according to (5). The extended internal energy maintains the vertex distribution of the corresponding instance of the PDM. The corresponding weights p_l are calculated as in Eq. (5) prior to the solution of the linear equation system (Section 2.1.4).

2.2.2. Model of spatial relationships between surfaces

In our previous work we considered segmenting single objects. However, the epi- and endocardium are two structures which are not connected but spatially related. During mesh adaptation, the spatial relationship may be violated due to missing features or attraction to false features. As a result, meshes can intersect or can be too close to each other. For example, the endocardial mesh can be attracted to epicardiac features in the image (Fig. 6(a) and (c)).

If a prior model of spatial configuration exists, it can be used to improve the robustness and accuracy of segmentation methods. For the segmentation of the cortex, Zeng et al. (1999) evolve two coupled geodesic surfaces simultaneously, advantageously integrating the cortical mantle thickness as a spatial constraint. The m-reps pioneered by Pizer et al. intrinsically enable to model multiple objects (Fletcher et al., 2002) and accommodate spatial relationships.

We propose a method for the simultaneous adaptation of connected deformable meshes. In order to approximately maintain the spatial relationships between object surfaces during deformation, additional vertex connections are introduced that connect neighbouring vertices of the existing endo- and epicardiac surface model (Fig. 2). Both surface meshes are adapted simultaneously while constraining their initial configura-

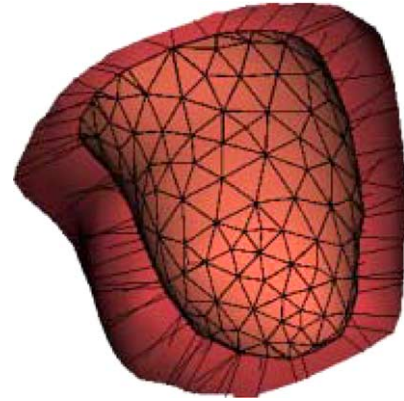


Fig. 2. Myocardial mesh with connections between endo- and epicardium (both cut in half for visualization purpose).

tion and the vertex connections by adding a connection energy E_{conn} to the deformation energy in Eq. (1)

$$E = E_{\text{ext}} + \alpha E_{\text{int}} + \beta E_{\text{conn}}, \quad E_{\text{conn}} = \sum_{j \in C} (\mathbf{v}_j^{\text{epi}} - \mathbf{v}_{k(j)}^{\text{endo}} - s\mathbf{R}\tilde{\mathbf{v}}_{jk})^2, \quad (8)$$

where $\tilde{\mathbf{v}}_{jk} = \mathbf{v}_j^{\text{epi}} - \mathbf{v}_{k(j)}^{\text{endo}}$ refers to a connected pair of vertices of the epi- and endocardiac mean surfaces, C is the set of connected vertices, β weighs the connection energy, and $E_{\text{int,ext}}$ are calculated as in (3) and (4) for both surfaces.

The connections are established once as part of the model construction phase (Section 3.3, Fig. 2). For each vertex in the epicardiac surface, the closest vertex in the endocardium is found in the sense of the Euclidean distance. Since the shortest distance is not symmetric, the same is done for the endocardium's vertices. The resulting set of vertex pairs is then reduced so that each vertex has at most one connection, where connections with smaller length are favored. This results in connections approximately orthogonal to the surface meshes in most cases, although orthogonality is not critical.

2.2.3. Spatially varying feature model

Specificity of image features is crucial for the robustness and accuracy of deformable model adaptation, in order to prevent attraction to boundaries of structures other than the object of interest. Deterministic, parameterized feature models have been successfully used, e.g. combining image gradient strength and direction w.r.t. the triangle normal, $F = \mathbf{n}^T \mathbf{g}$ for vertebrae in CT. However, the gradient direction varies across the myocardiac surface, and the gradient strength alone is not a robust feature due to neighbouring structures with similar contrast (Section 3.4, Fig. 6).

Statistical feature learning methods have been applied to construct optimal feature functions. Statistical profile matching based on training data (Cootes et al., 1998; Mitchell et al., 2001) models variable grey value appear-

ance which is mainly due to the variability in the training sample and incorrect ground truth. Profile matching during adaptation is computationally expensive in 3D and achieves low specificity if the inter-subject profile variance is high. Fig. 3 shows the grey value profiles taken along the normals of corresponding triangle centers at two particular locations of the mesh. While the left example is suitable for a univariate Gaussian model, the right profile shows a multivariate distribution of location of gradient maximum and boundary function (i.e. edge, ridge, etc.), requiring a more complex statistical model.

We propose a simpler, hybrid approach, to make feature search more robust to outliers. The parameters of a deterministic feature model are learned once from a training sample as part of the model construction. Specifically, for each triangle j we learn a grey value range $I_{\min}^j, I_{\max}^j$, a gradient range $G_{\min}^j, G_{\max}^j$, and a gradient orientation $s_j = \pm 1$ w.r.t. the triangle normal (Fig. 4(b)) corresponding to the intensity transition (“dark–bright” or “bright–dark”) at the object surface. This set of surface characteristics can be regarded as a specific common denominator of myocardial surface features.

Thus, the image feature $F_j(\mathbf{x})$ for a triangle j in (2) is defined as

$$F_j(\mathbf{x}) = \begin{cases} s_j \mathbf{n}^T \mathbf{g} & \text{if } |\mathbf{g}| \in [G_{\min}^j, G_{\max}^j] \text{ and } I(\mathbf{x}) \in [I_{\min}^j, I_{\max}^j], \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

The parameters of F_j are automatically determined using the corresponding triangular meshes generated above (Section 2.2.1). First, grey value profiles are sampled in all L images at all T triangle barycenters along the triangle normal. Then all triangles are assigned a particular tissue interface class $c_{j,l}$, $j=1 \dots T$, $l=1, \dots, L$, using square-error (“k-means”) classification (Duda and Hart, 1973) on the profiles. We chose the $C=3$ classes myocard-fat, myocard-right ventricle, and myocard-lung parenchyma. The triangle class memberships were initialized so that each class contained an equal number

of triangles. The sample size for classification is approximately $L \cdot T/C$ as opposed to L for local profile statistics, which makes it more robust against outliers.

After classification, the T triangles of the shape model are labeled with class label C_j and assigned the corresponding s_j according to

$$C_j = \arg \max_{c=0,1,2} \sum_{l=1}^L \mathbf{1}_{c_{j,l}=c}, \quad (10)$$

based on the property that the triangles j, l have approximately anatomically corresponding locations on the learning examples. Similarly, the triangle-specific grey value and gradient value ranges are the means $\pm$ (std deviation) of grey values and gradients in the corresponding triangle barycenters in the learning set.

3. Experimental results

For 3D adaptation of the shape model, automatic initial positioning of the shape model in the center of the image volume was sufficient for the cases in our image database. In cardiac imaging, the location of the heart can be assumed approximately in the center of the image. To increase the capture range of our method, we adopt a coarse-to-fine adaptation approach on an image resolution pyramid generated with a fast B-spline based method (Lee et al., 1998). Because the in-plane resolution of the images is significantly higher than in the slice direction, resampling is only done in-plane, using two lower image resolutions, i.e. 128×128 and 64×64 in-plane resolution. Computation time for the resolution pyramid was in the order of 5–6 ms. Nevertheless, in case of an unfavorable field of view, more sophisticated positioning is possible but was not necessary in this work (Mitchell et al., 2001).

The parameter settings were chosen experimentally, Table 1 summarizes the settings for the adaptation to grey value images. It was previously shown that the

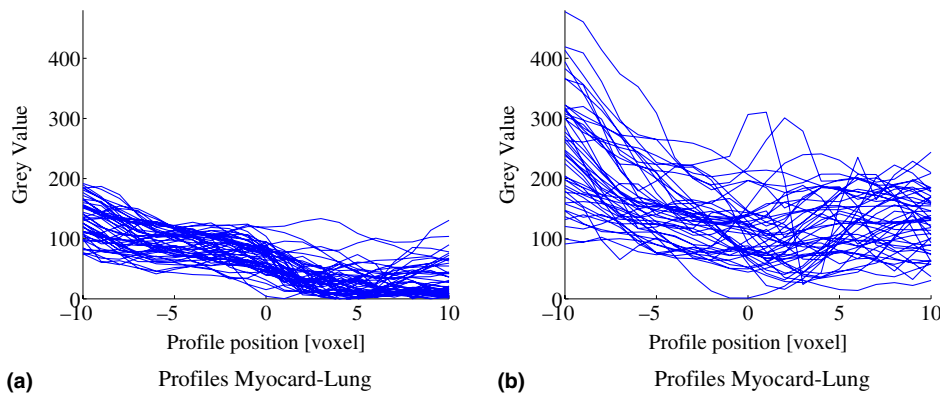


Fig. 3. Depending on the triangle location, profile statistics may exhibit small (a) and large (b) variations, drastically reducing the specificity of profile matching.

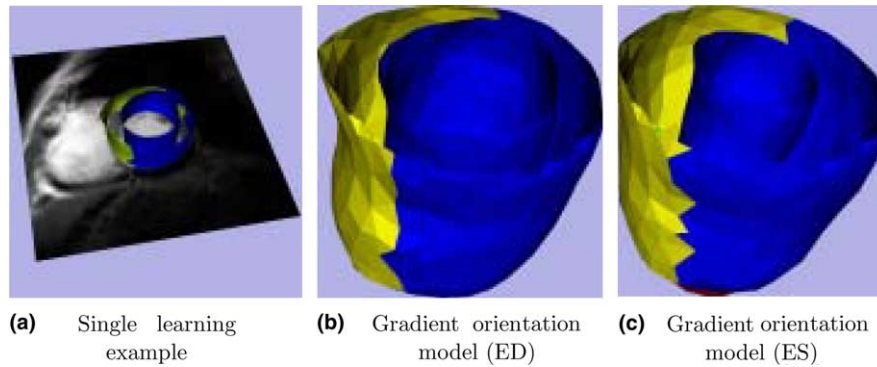


Fig. 4. Feature model accommodating the spatially varying surface characteristics learned from the training database. The triangles are labeled according to the tissue interface (color-encoded). As expected, the tissue class distribution changes between phases (end-diastolic, -systolic) are small and in radial direction.

adaptation method is robust w.r.t. different parameter settings when used for shape model generation (Kaus et al., 2003), and behaves comparably robust when used for grey value images.

The adaptation to a 3D image took approximately 2 s on a single-processor 2.7 GHz Pentium 4, such that the overall adaptation procedure including three mesh adaptations to the image pyramid amounted to approximately 7 s.

3.1. Image data

Our test database consisted of cardiac MRI datasets (ECG-triggered FFE-EPI, Philips ACS-NT 1.5 T, 20

time frames, 256×256 images with 7–10 slices, $1.25 \times 1.25 \times 10$ mm) from 169 cases with undetermined pathology. In all cases, the myocardium was manually segmented in the end-diastolic (ED) and the end-systolic (ES) phase. We used a dedicated, simple tool where points were selected slice-by-slice on the boundary of the epi- and endocardium on the short-axis views. The points were connected to a closed contour with B-splines and a labeled image was derived by filling the contours.

Some datasets exhibited significant acquisition artifacts, e.g. due to breathing and through-plane motion (Fig. 5). We excluded those cases from our study because they cannot be represented by a smooth shape and would therefore bias the overall error statistic. We do not address this problem, mainly because more recent, faster cardiac MRI protocols like, e.g., balanced FFE, do not pose this problem. Unfortunately, we do not have a sufficiently large segmented database of more recent images. In total, we used a subset of 121 subjects.

3.2. Segmentation quality measures

We compared the automated results to the manual segmentations by calculating the mean and maximum surface distances between the mesh and the binary im-

Table 1
Parameter settings for adapting meshes to grey value images

Parameter	Value
Energy weight, α	0.25
Distance weight, d	100
Profile step size, h	1 mm
Profile length, l	10
Number of eigenmodes $\tilde{L}$	14
Mesh reconfiguration steps	40

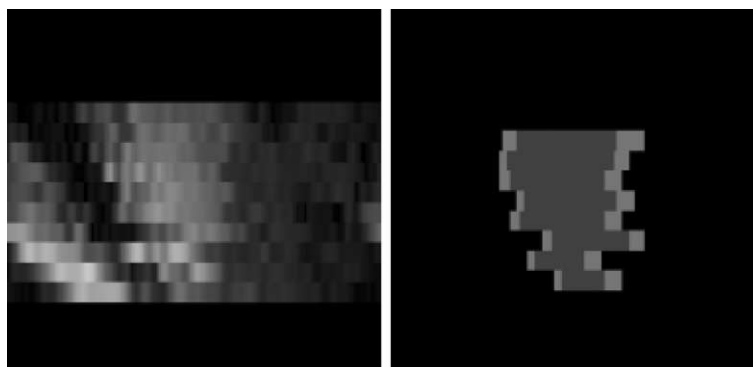


Fig. 5. Studies with significant shifts between slices due to motion artifacts were excluded from the database.

age of the manual segmentation. This was done by (i) extracting the surface voxels of the epi- and endocard in the segmented images by selecting those voxels which are 6-connected to the background, (ii) computing an Euclidean distance transform and (iii) evaluating the distance values at the triangle centers of the adapted mesh.

To evaluate the 3D adaptation, we conducted a leave-one-out experiment, where the prior model was built from all images but one, and adapted to the image left out. This procedure was repeated for all images.

3.3. Model construction

We constructed two organ models, one for the ES and one for the ED phase. First, we derived separate triangulations for the endocard and epicard of a segmented im-

age in the end-diastolic phase. In accordance to clinical practice, the FOV was defined to include the entire left and right ventricle from the apex to the valve plane. Therefore we cut both meshes open using VTK (Schroeder et al., 1996) because this part of the surface has no corresponding object boundary in the image. Subsequently, the connections between the two open surfaces were established, resulting in the shape template (Fig. 2).

The shape template was then adapted to all remaining 120 segmented images for both phases. This resulted in two PDMs with corresponding vertices (see Table 2 for the PDM characteristic). The end-diastolic LV shape model reached 95% (98%) variance with 5 (11) eigenmodes, and the end-systolic LV shape model 95% (98%) variance with 6 (14) eigenmodes, so that we chose to use 14 eigenmodes throughout the adaptation process. This coincides with visual inspection, where the

Table 2

Statistics of the shape model generation process: surface errors between shape template adapted to the segmented images during model construction and the object surface (endocard/epicard)

Adaptation error (mm)	Mean	SD	Max
End-diastole	1.79/2.12	0.31/0.36	4.62/5.95
End-systole	2.18/2.03	0.55/0.48	5.88/5.64

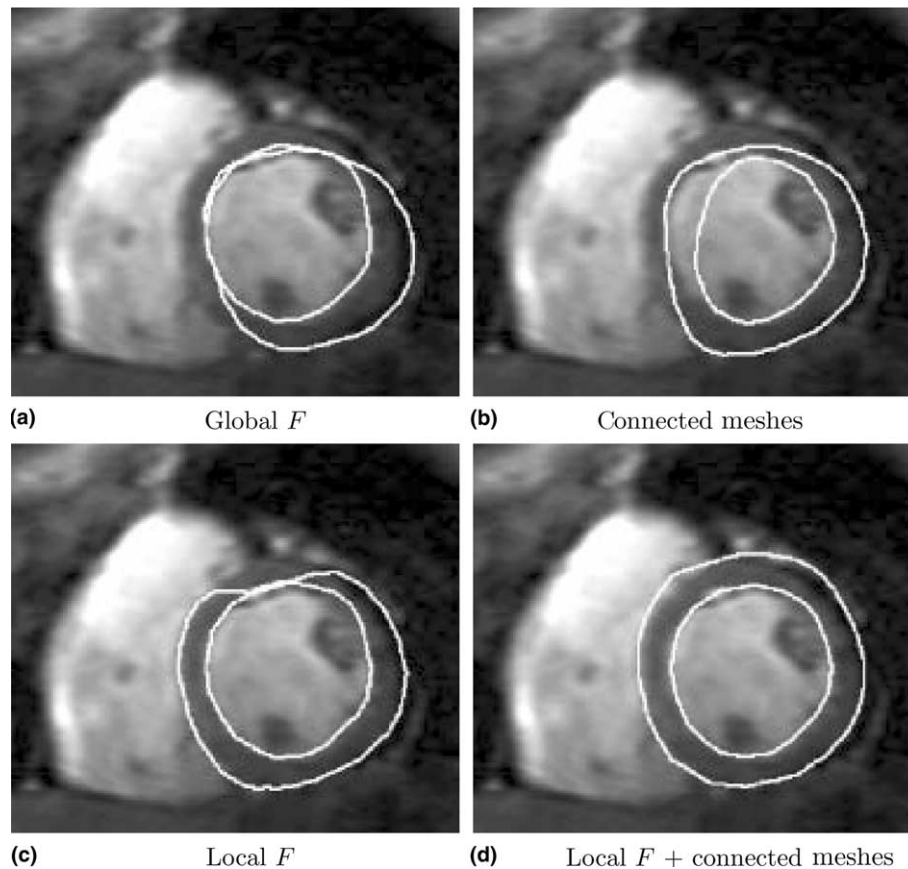


Fig. 6. Effect of prior knowledge on model adaptation. F is the feature function.

inter-subject variability of the myocardial thickness is larger in end-systolic than end-diastolic phase. No significant qualitative difference was observed between ES and ED model, because the data source (segmented images) and the object complexity is comparable.

Next, two feature models were generated separately for ED and ES (Fig. 4). The main measured difference was a slight rotation of the entire tissue interface distribution, which is due to the heart movement. Thus, learning a feature model over the entire phase would result in smeared boundaries between areas of tissue interfaces on the surface, thus we sample a feature model only from a single phase.

3.4. 3D adaptation

We illustrate the effect of the multi-model knowledge sources with an example in end-diastoly. Separate adaptation of the two surface meshes with a global feature function $F = \mathbf{n}^T \mathbf{g}$ (Fig. 6(a)) results in severe adaptation errors since it models a bright object on a dark background. Simultaneous adaptation of connected meshes approximately maintains the distances between the meshes (Fig. 6(b)), but does not prevent attraction of the epicardiac mesh to the LV. Separate adaptation with a spatially varying feature model (Fig. 6(c)) correctly adapts the epicardiac mesh in places where the myocar-

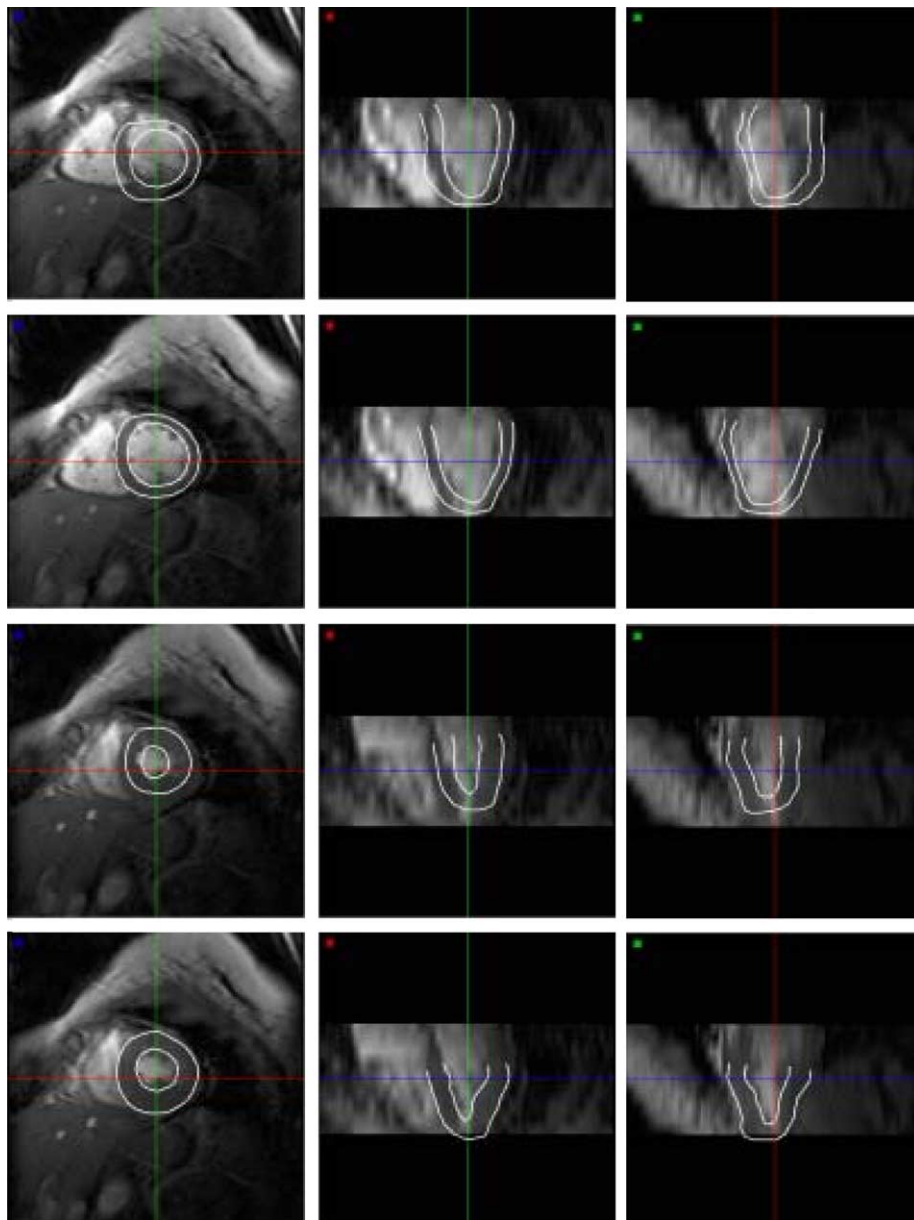


Fig. 7. After initialization of the end-diastolic (1st row) and end-systolic model (3rd row) in the center of the image volume, automated adaptation (2nd and 4th row) achieves visually satisfactory results.

Table 3

Adaptation errors in the end-diastolic and end-systolic phase (endocard/epicard) after initial positioning of the mesh in the center of the image volume, adaptation using a gradient strength feature, and the combination of multi-model prior information

Adaptation error (mm)	Mean	SD	Max
<i>ED</i>			
Initialization	2.47/3.73	1.87/1.93	20.92/21.88
Constant gradient	3.67/4.32	1.93/1.13	21.76/17.68
Multi-model Combination	2.28/2.62	0.93/0.75	13.82/12.35
<i>ES</i>			
Initialization	3.05/4.02	2.11/2.35	22.83/22.97
Constant gradient	3.84/4.58	2.32/2.73	20.95/23.58
Multi-model Combination	2.76/2.92	1.02/1.38	15.43/17.67

diac tissue interfaces with the right ventricle. Nevertheless, the partial differences between the feature model and this subject result in false attraction to the epicardiac fat. Constraining the spatial relationship of the surface meshes (Fig. 6(d)) compensates the partially wrong feature model. For wall thickness analysis the papillary muscles are considered to be inside the endocardiac surface, which is achieved by maintaining the shape of the model and the spatial relationship between the meshes.

Fig. 7 shows an example of the segmentation results in three orthogonal image planes from the end-diastolic and end-systolic phase of a subject. The visually satisfactory adaptation results are confirmed with our quantitative analysis (Table 3). Accurate adaptation is not possible using shape constrained models with the simple gradient strength feature. Mean and maximum errors after automated initialization and mesh adaptation are 2.45 ± 0.75 mm (2.84 ± 1.05 mm) or 1–2 voxels for end-diastole (end-systole), see Table 3. The maximum errors occur mainly in the direction of the 10 mm slices. The adaptation to the endocard performed slightly better than the epicard adaptation. This can be explained with the high variability of tissue interface distribution across the surface of the epicard, while the endocard only interfaces with the myocardium.

In some cases the initial mesh was a better approximation to the subject's LV than the adaptation result. This was mostly due to the epicardial mesh adaptation errors as described above. They can be explained with the spatial distribution of features in the model that did not describe the subject's grey value appearance well and could not be compensated by the shape constraints (i.e. surface and connections), resulting in partial attraction of the endocardiac mesh to epicardiac fat.

The adaptation results for ED were closer to the manual segmentations than for ES. This can be explained with the significantly higher inter-individual shape differences, mainly in the muscle thickness of the apex (Fig. 8).

The adaptation errors to segmented images (Table 2) represent the best case in terms of image features, and

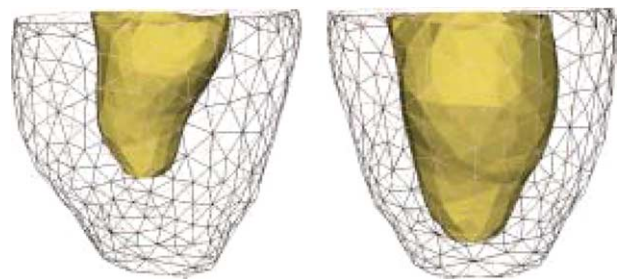


Fig. 8. The endocardium (surface) and epicardium (wire frame) meshes after adaptation to two different subjects. Significant inter-individual shape differences can occur in the area of the apex in the end-systolic phase.

thus are also a measure for the lower bound of our method to represent objects in 3D images. This representation error also explains a part of the grey value adaptation errors in Table 3.

4. Conclusion

We presented a method for modeling and fully automated segmentation of the LV in 3D cardiac MRI time-series with a mean error of 2.45 ± 0.75 mm (2.84 ± 1.05 mm) for the end-diastolic (end-systolic) phase. The method is fast and requires approximately 7 s on standard hardware for the segmentation of a heart phase.

We demonstrated that the integration of multi-model prior knowledge into a deformable model is possible in an efficient way, and significantly increases the robustness and the accuracy of segmentation with such techniques. Information about shape and spatial configuration of objects by simultaneous adaptation of coupled meshes helps compensating the lack of prominent image features. Learning of parameterized, deterministic local feature functions enables robust surface detection for objects with spatially varying grey value appearance on the surface, and an explicit 4D deformation model enabled to segment objects of large shape variation.

Several interesting topics for future work remain. We need to investigate the applicability of our framework to

higher quality MRI and also other modalities, e.g. CT, which has the advantage of higher resolution but involves the development of an appropriate feature model. The extension of our method to multiple objects to include, e.g., the right ventricle and other parts of the heart is another important topic. Finally, the investigation of using spatio-temporal information, e.g., in a tracking approach, to segment all time-phases and enable the extraction of motion specific parameters is currently being investigated.

Acknowledgements

Cardiac MRI have been provided by Warren J. Manning, MD, Co-Director of the Cardiac MR Center of the Beth Israel Deaconess Medical Center, Boston, MA. We thank Miriam Siers, Image Sciences Institute, University Medical Center, for the manual segmentation of the cardiac MRI. We thank our colleagues C. Lorenz, T. Netsch, K. Meetz, Philips Research Hamburg, and S. Lobregt, R. Truyen, F. Gerritsen, Philips Medical Systems Best, for useful discussions. M.R.K. thanks D. Rückert, Imperial College London, and B.P.F. Lelieveldt, Leiden University, for helpful comments on the original MICCAI conference article.

References

- Cootes, T., Taylor, C., 1995. Combining point distribution models with shape models based on finite element analysis. *Image Vision Comput.* 13 (5), 403–410.
- Cootes, T., Edwards, G., Taylor, C., 1998. Active appearance models. *Proc. ECCV. LNCS*, vol. 1406. Springer, Berlin, pp. 484–498.
- Duda, R., Hart, P., 1973. *Pattern Classification and Scene Analysis*. Wiley, New York.
- Fletcher, P., Pizer, S., Gash, A., Joshi, S., 2002. Deformable m-rep segmentation of object complexes. In: *Proc. ISBI*. IEEE Press, New York, pp. 26–29.
- Frangi, A., Niessen, W., Viergever, M., 2001. 3D modeling for functional analysis of cardiac images: a review. *IEEE Trans. Med. Imaging* 20 (1), 2–25.
- Gill, P., Murray, W., Wright, M., 1981. *Practical Optimization*. Academic Press, San Diego, USA.
- Golub, G., van Loan, C., 1996. *Matrix Computations*, third ed. John's Hopkins University Press, Baltimore, MD, USA.
- Jolly, M.-P., Duta, N., Funka-Lea, G., 2001. Segmentation of the left ventricle in cardiac MR images. In: *Proc. ICCV*. IEEE Press, New York, pp. 501–508.
- Kass, M., Witkin, A., Terzopoulos, D., 1988. Snakes: Active contour models. *Int. J. Comput. Vision* 1 (4), 321–331.
- Kaus, M.R., Pekar, V., Lorenz, C., Truyen, R., Lobregt, S., Weese, J., 2003. Automated 3D PDM construction from segmented images using deformable models. *IEEE Trans. Med. Imaging* 22 (8), 1005–1013.
- Krahnstöver, N., Lorenz, C., 2004. Computing curvature-adaptive surface triangulations of three-dimensional image data. *Visual Comput.* 1 (20), 17–36.
- Lee, C., Eden, M., Unser, M., 1998. High-quality image resizing using oblique projection operators. *IEEE Trans. Image Process.* 7 (5), 679–692.
- McInerney, T., Terzopoulos, D., 1996. Deformable models in medical image analysis: a survey. *Med. Image Anal.* 1 (2), 91–108.
- Mitchell, S.C., Lelieveldt, B.P.F., van der Geest, R.J., Bosch, H.G., Reiber, J.H.C., Sonka, M., 2001. Multistage hybrid active appearance model matching: segmentation of left and right ventricles in cardiac MR images. *IEEE Trans. Med. Imaging* 20 (5), 415–423.
- Lorenzo-Valdes, M., Sanchez-Ortiz, G., Mohiaddin, R., Rückert, D., 2002. Atlas-based segmentation and tracking of 3D cardiac MR images using non-rigid registration. *Proc. MICCAI. LNCS*, vol. 2488. Springer, Berlin, pp. 642–650.
- Montagnat, J., Delingette, H., 1998. Globally constrained deformable models for 3D object reconstruction. *Signal Process.* 71 (2), 173–186.
- Montagnat, J., Delingette, H., 2000. Space and time shape constrained deformable surfaces for 4D medical image segmentation. *Proc. MICCAI. LNCS*, vol. 1935. Springer, Berlin, pp. 196–205.
- Pentland, A., Sclaroff, S., 1991. Closed-form solutions for physically based shape modeling and recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* 13 (7), 715–729.
- Schroeder, W., Martin, K., Lorensen, B., 1996. *The Visualization toolkit: an object-oriented approach to 3D graphics*. Prentice-Hall, Englewood Cliffs, NJ, USA.
- Weese, J., Kaus, M.R., Lorenz, C., Lobregt, S., Truyen, R., Pekar, V., 2001. Shape constrained deformable models for 3D medical image segmentation. *Proc. IPMI. LNCS*, vol. 2082. Springer, Berlin, pp. 380–387.
- Zeng, X., Staib, L.H., Schultz, R.T., Duncan, J.S., 1999. Segmentation and measurement of the cortex from 3D MR images using coupled surfaces propagation. *IEEE Trans. Med. Imaging* 18 (10), 927–937.